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LOGICAL FICTIONS.

I. WHAT WE KNOW AND HOW WE KNOW IT.

I.

IT is not easy to say what it is we know and how we get to know it: to do so, we must first face other questions which in our ignorance we answer in various ways. Our chief foe is common sense that takes so much for granted. Many take the common-sense view that just as water is water, so thoughts are thoughts and knowledge knowledge. That's good enough for them; they do not feel the need for more.

But a few feel they must look more closely. In all ages, perhaps, men have tried (without much success) to analyze "knowledge" and to understand how we get it; but in recent years progress has been so considerable that any one interested in the subject now starts with the great advantage of having a large number of obstacles removed: many of the prejudices of common sense have gone forever, and, to some extent at least, the way to an answer has been pegged out.

Any one who wishes to reach a satisfactory explanation has to do some strenuous thinking for himself; but that the goal is (if it is) now almost in sight, is due to the labors of many hundreds.

But if a writer on this subject must have a strenuous time, so must the reader, for the writer may take points of view for granted that are new to the reader, and it is always difficult to follow an argument in which too much is taken

for granted. In order to prevent this as far as possible, it will be necessary to solve some of the more important difficulties offhand and dogmatically and to start with certain assumptions, leaving detailed treatment for later chapters. Just as it is not possible to do much geometry without accepting some definition of point and line, so it is not possible to get far in epistemology without *starting* with certain assumptions; it will be seen later that the only assumption we have to make is that *it is necessary to start*.

For the present we must use the common-sense phrase "to use one's senses." We cannot, of course, leave it at that, but it will do for the moment.

"Space" and "time" in the common-sense meaning are as irrelevant to our problem as a sheet of paper to the sum that is done on it. It is important to realize from the outset that common-sense ideas of time and space are merely useful habits that express in a crude way certain subtle differences in what we call "here" and "there": if used for analysis they are grossly misleading. It is, for instance, misleading to speak of the present as if it were time at all like past and future: it is only the locus of abstracts that do not exist in time. The future and the past are two aspects of the same time which is the locus of all instances of numbers (*inter alia*). The moment these instances are recognized as individuals they are localized in space.

Common sense misleads us even in the use of our senses. It is hard, as a rule, to get ourselves to look at what we see; usually we look only at what we think we see—a different thing altogether. When we look out of a window, we fancy we see things at various distances, but we don't. We see nothing but many colors and shades.

Again: most of us think that we are bound to see what we think we see. There may be cases, we admit, where we "see wrong"; but we feel that the thing was there to see all the time. It is a little hard to persuade ourselves that

we see things only as we choose to see them. It is folly (but, practically speaking, a very necessary folly) to imagine that there is one way in which things must be seen: for if we look at a sheet of paper with a black disk in the center, it is obvious that you can see it either as a dark globe in front of the paper or as darkness behind seen through a hole in the paper. Any one will admit that there may be cases where we cannot say for certain what it is we see. We cannot always say whether a cloud is behind or in front of another. But we all find it hard to believe that the more we have "a good look all round it" and touch it and move it, the further do we get from what we see, because we are mentally constructing a complete picture which is invisible except to the mind's eye.

Even from very early years of life we cease to see things as they appear. An easy way of seeing this is to look at the floor: if we *only* look at it, it seems to do the most ridiculous things.

If, then, things can thus be seen in various ways, what is it makes us decide one way or another? The difficulty of this question will not be appreciated unless it is recognized that we can never see anything completely: what we see is a part (often an irrelevant part) which acts as a symbol to suggest the rest. There is no real difference between seeing a circular black spot either as a disk or globe on this side of the paper or as darkness beyond; seeing the sign "I" as a figure on this side or as darkness beyond; seeing "I" as a symbol of self. In all these cases, what we see is a mere fragment which suggests a great deal that is "not there." It is no exaggeration to say that all these symbols open a window on an infinity. It is the same when I recognize you or your dog or your house. In no case do I see the whole of what I mean by you, your dog, or your house. But something or other calls up what I mean by you or your dog or your house. It is quite possible I might

recognize you all by your shadows, just as I can recognize you by words that stand for you, or photos. We admit that pictures are symbols; we hesitate to admit that what our senses receive is a symbol and has been ever since early childhood.

A symbol for what? For what we put there. If we were standing outside a room and heard sounds coming from within—voices, dishes, footsteps, glasses, knives and forks—we could build up a mental picture of what might be going on. Then if we opened the door, we should not be in the least surprised to find that we had not got the picture quite right. Our eye would at once modify the picture, and we should then have a new one which we foolishly suppose to be final. For we do feel surprised now, if told that this new picture is no more likely to be final than the other. In fact, so little finality is there in it, that the longer we look the more we change it, because we notice something new each moment. Even granting that we could at any moment see such a picture in “final shape” as it “really is”—would it stay as such more than a moment? So engrossed are we with our mental construction (which is comparatively easy to fix) that we notice only with great effort that the “solid fact” which I call your house looks so different at different times of the day and from different points of view, that it would be quite unrecognizable unless treated as a symbol: we recognize your ever-changing house as easily as we recognize the word house, however carelessly or quaintly written.

It is open to any one to believe that he has “pure sensations”: it is unnecessary (and certainly inadvisable) to assert that we never have pure sensations. But it is certain that the instinct to “explain” sensations works so keenly that it is very difficult to distinguish a sensation from the “explanation” that follows it. Sensations seem to bounce, and as soon as they bounce they have become

explanations. Any light, sound, or pain that cannot turn into an explanation terrifies us. And because we all explain, we think that we explain rightly—it is hard to convince ourselves that the function of an explanation is not to explain but to satisfy. Neither peasant nor scientist can explain light; both find a solution that satisfies them; if a peasant finds it easy it is because he is easily satisfied. When an apple falls from a tree, is it natural? is it weird? You may say “Of course it falls—what else should it do?”—or “gravitation, of course”—or anything else you like. It is weird; but for the sake of sanity, you must explain away the weirdness, so that it seems natural and you feel at home, master in your own world.

If, then, we all read the face of the world just as we read a book, seeing symbols and by their means making a mental construction which we call reality; and if we all instinctively veil as soon as possible anything mysterious or inexplicable with an explanation in order to feel masters of the situation: do we have to learn to read the external world just as we learn to read a book? Obviously; and, fortunately, when we are very young we have plenty of time.

The question remains: How do we learn to read at all? It seems such a natural, homely thing that most of us are satisfied with the “explanation”: “Oh, well, we all just learn.” It is extremely difficult to explain how we come to know for instance what the word *dog* stands for. We all imagine that we see hundreds of instances of *dog*. But who ever “saw” what *dog* stands for? All visible dogs are symbols: what *dog* stands for is invisible even to the mind’s eye; “a dog,” being only an instance, is visible only to the mind’s eye; and this dog, Fido, is a group of symbols visible to the eyes.

There is a matter here which it is important to deal with: we find it hard to get rid of the idea that behind some

symbols there is what is called "the real thing." We are apt to think that we have, as it were, a scale, starting from symbols that are what may be called "exactly like," to symbols, like a printed word, which are "nothing like." And if a living dog, that we see, is a group of symbols, why are some symbols more "real" than others? It will be easier to see this difference if we first make sure of what we mean by counting. It is commonly supposed that it is possible to count things—trees, tables, coins, and so on. And not only that; we are supposed to be able to count them by merely looking at them or "ticking them off." Yet it is easy to find two tables, for instance, that have *absolutely nothing* in common, except a very-hard-to-define tableness. A table is a table because it answers to your (or some one else's) idea of a table. There can be no doubt whatever that we never count "things": we count instances of thingness. If I say I see a house, what I *see* is a group of symbols which stimulates my idea of house. What I *think* of is, (1) house, (2) a house, (3) that house. Thought No. 3 is equivalent to a proper name for what exists in space. Thought No. 2 is of what exists only in time (past or future according to your point of view): it is an instance not only of "house" but of "one." Thought No. 1 is of what does not exist in time but only in the present, being, like what "one" stands for, an organized (perhaps it would be safe to say an organic, growing) abstract.

Now, what is this thingness—this organized abstract—that we can count when we come across instances of it? A pattern. Any object seen, anything heard, touched, tasted, or smelled is distinguished by a characteristic pattern. The external world is, let us say, nature's writing. If we see written the word "Bill," it has, we say, a number of meanings. For a man whose name happens to be William—whose father and eldest son are also William—the word Bill is likely to be a very live symbol, even if it had

no other meanings. And the only reason why a "real" dog seems more real than (say) a picture, is because the real dog is a group of symbols that stands for more than merely "dog": it answers to a number of other very mysterious but quite obvious "thingnesses," like "life," "animal," "dangerous," and so on, to which the printed symbol does not answer.

We have, then, primarily an endless collection of symbols, and just as our hands can grip a handful which varies according to the matter gripped, so the mind can grip a pattern by which we prove experiences. If any experience proves to be an instance of the pattern, we can count it. By counting it, we have shifted it from an abstract present to actual time (past or future). We may then give the instance a name and place it in space.

Every pattern has, or could have, a name. The external world is not (so far as we know) full of things: for us, it is full of instances of patterns—most instances being parts of many other patterns. There are infinitely more patterns than instances—many patterns cannot even have one instance.

A pattern must not be thought of as something fixed. A pattern survives constant change by means of something very like personality: it is hard to say which is the more slippery and shifting, what the symbol "Bolshevist" stands for, or what instance will call for that description. In fact, one is tempted to say that the personality of the pattern is only a complicated relation: just as there can be no such thing as "angle" without its limits, so a pattern like Bolshevism is formed and contained, and like an angle it can have degrees. We all experience the sameness and yet the difference of a pattern: the first view may give us what is called "a totally wrong impression"—say, of Pussyfoot. But the important thing is not the "totally wrong" but the impression, which being abstract, exists outside time, that is,

permanently in the present. When we feel that the impression was totally wrong, it merely means that the pattern is like a swiftly growing crystal, or like capital at compound interest.

However difficult it may be to see clearly what a pattern is, it is easy to recognize an instance of a pattern. Who first thought of stimulating thought by symbols in imitation of nature, we do not know. Nature's symbols are essentially the same as the artificial ones we use (for instance) in language. If we read *wind* in a book, we know that we have only a symbol before us. Why is it that, when we read the symbol for the same pattern in nature, we imagine that we are doing more than reading symbols—and adding explanations?

These explanations must now be dealt with more fully and will be the subject of the following section.

II.

Before trying to reduce the number of assumptions we have made, let us repeat three important points.

1. Knowledge is founded on consciousness, which it is for physiologists to explain. We start with being aware. We start by being aware of something: before that, we were aware of nothing; since then we have never been (and can never be) aware of nothing—we can only be aware of something or something else. When we say that there is nothing in a box, we are merely stating that there is something else—not what we were thinking of. “Awareness” is, as we have said, for physiologists to explain—it seems to be similar to what is called chemical reaction or tropism. But there are various kinds of awareness: light is beautiful but it may hurt. It seems safe to say that normal human beings like light and hate being burnt. Knowledge depends not on being aware of the light or of the burn, but on connecting the two in one pattern. Like much else in nature,

this combining process seems to be a matter of chance and habit. Once the pattern is complete we wonder how people could have failed to see the connection: we now easily connect death with lightning rather than thunder. If, then, knowledge is founded on consciousness, it is built up on connections. Common sense is so misleading because it is full of false connections. Blankets keep you warm, therefore it must be absurd to wear them in a very hot climate. When the sun shines in a clear sky the heaven is full of light, therefore on the surface of the moon the sky couldn't look black if the sun were visible. And so on. By correcting connections and complicating them in the light of experience we advance in knowledge. A sum of such connections is what is represented in a word.

2. Mere awareness, say of light or a burn, seems to be similar to what a steel pen "feels" in the presence of the magnet. There seems to be a wide range of life that survives without more than awareness: food is accepted, and either digested or rejected. Knowledge starts when it is possible to satisfy ourself without trial. And it is only possible to do this if the sensations or feelings stimulate a connection or pattern, and not merely a reaction. An instance of this would be this: I see a bird on the wing and I want to shoot it; what I shoot at (if I am to hit it) is not the bird but a "pattern" which I form by means of a most intricate set of connections.

3. We are not here concerned with the origin of this intricate set of connections. For the present it is enough to see clearly that sensations in the widest sense of the word stimulate a reaction which requires an explanation to satisfy it, or, as we might say, to explain it away. We are held up by what we cannot explain away. For instance, in a battle the whistling of bullets arrests, until we "know" it: a man falls dead—a rumbling sound—flashes—utter stillness—lights at night—all these we can ignore, only if

we can give an explanation of them. The explanation may be quite wrong, but provided that it satisfies us, we call it knowledge. If we say that a solar eclipse is due to a dragon and the explanation satisfies—that is knowledge. The moment it ceases to satisfy, we require another explanation.

We shall now be able to see that all explanations are given in terms of motion: the fundamental explanation is roughly equivalent to what we call a present participle. "Something" is essentially thought of as being capable of being distinguished or *separated* from "everything"; a thing cannot be thought of otherwise than as "*being a thing*"; no thing can be thought of as permanent in time, but it can be thought of as permanent in the "present," but only because that is a timeless abstraction of permanent change. This permanent pattern which cannot exist in time is and must be thought of as something "picked out": the moment you see a pattern, you have got something which stands out; in other words, the rest—everything else—falls away from it. The pattern of the Great Bear is got only by thinking of those stars as being on a plane of their own. Once this property of "being" is grasped, it is hard to resist the idea that all "knowledge" is due to the fact that we move and that things move. How we perceive motion is a problem with which we are not concerned. The common-sense idea is that we see it—but it seems likely that we shall not have to wait long before it is proved that, although the eye obviously supplies a large amount of the matter to be read, it is the ear that actually reads the motion.

The process can be expressed symbolically. Thus, if a stands for something perceived, the next step is represented by $a-b$, indicating that we have thought of b as moved from a . Thus, if we perceive a tree (b) it is clearly taken out of or thought of as separable from everything else (a). If we take one step further we shall be able to

show that if we *add* to our perception of the tree we are still subtracting something from the whole. If we notice the branches and call them c , we have clearly got as our pattern $b + c$: we have subtracted $b + c$ from a . Thus it is clear we can state the process either as $a - (b + c)$ or as $a - b - c$.

Common sense would say that knowledge depends on the number of terms of the series $a - b - c \dots - n$. The longer the better. The contrary is true. It does not even require God—any decent human brain can be trained to see that

$$a - (b + c + d \dots + n) = a.$$

For practical purposes that is of no use: we do not require explanations of that kind, we want explanations that will satisfy expectations. What "expectations" are, we must again ask the physiologists to explain: it is enough to suggest that "expectation" may be one (if not *the*) characteristic of "life." And the only explanations that satisfy expectations are these: if I perceive a pattern c , I expect that pattern to do its duty; if I come across another c , I not only expect it to be a c , but to do its duty. I may be extremely vague as to what its duty is, or I may be quite wrong; but the moment I have the pattern at all, I have some idea: the event may satisfy me or surprise me. If it satisfies me I have the pleasant "I-told-you-so" feeling; if it surprises me I am nonplused and my series $a - (b + c \dots)$ undergoes revision. For instance: as a child, I get the pattern "star" by seeing the night sky and a strange plump object with five or six sharp points in picture-books; at school, I have to learn by heart something that says the stars "sang for joy"; later I am taken to the theater and see other stars singing for joy; and so on. Throughout, as soon as "star" means anything, we expect every star to do its duty. For practical purposes, then, it

is evident that what we want is neither a great nor a small number of patterns; wise is the man whose patterns not only are expected to do their duty, but do it.

The $a-b$ process is not hard to visualize. If a is the ocean and b is a drop, it is clear that b existed before it was separated from the ocean, but it did not exist as a drop. Likewise a thing existed, still exists, and will continue to exist, as far as we can tell, for ever or for no time in the "ocean," but it cannot exist as a thing until we recognize it as a pattern. No doubt the tiger existed in the Garden of Eden, but the tiger only became a tiger when Adam saw it and named it. What it was before that, God only knows.

Or, again, a pattern is like a pearl: it has a beginning—something which starts it—stimulates the formation. In time successive layers are added, it is the same pearl but a growing one in which the old surfaces become invisible. Even common sense will admit that my idea of a bath-bun changes as I grow older. What common sense refuses to admit is that, strictly speaking, nothing ever is quite the same as it was, because every experience is bound in some small (perhaps imperceptible) way to modify "the duty we expect."

If, then, every conceivable "thing" is thought of essentially in the form "being a thing"—much as we speak of "a saying," "a soaking," "lightning"—"thing" means a pattern, and "being" in this sense implies existence *as a pattern*. What we ordinarily call a "thing" exists in some form that is not recognized, just as a drop in the ocean is not recognizable. Of course, it is possible to think of patterns that are unsuccessful. As children we might conceive "a tree with clouds in its branches"; as such, that pattern will have its duty to perform: it is undoubtedly an explanation of what is seen. But sooner or later this explanation will cease to satisfy, and explanations that do not satisfy

are called wrong. It is important to admit that, although it is *useful* to believe that right is right, right is really only an explanation that has not yet been found to be unsatisfying or "wrong."

Perhaps the easiest way of getting a clear idea of this "pattern" is to think of it as a diamond. "Cat" is a pattern that comes early in life, and we think we know a cat because it looks like one. We don't. We no more know a cat because it looks like one than we know a rough diamond because it looks like one. In both cases we tell by what is done, i. e., the duty we expect. Let us take only one duty: the diamond if cut will flash, the cat if stroked will purr. Other things flash and other things purr, but a diamond flashes in its own way and a cat purrs in its own way. And the interesting thing to notice is that if there were no light we could not recognize the "duty"; and if there were no surroundings or circumstances we could tell no duty. It seems that the duty is always in the circumstances and never in the pattern, just as what we see in a looking-glass is never in the looking-glass. That is why we said that what *dog* stands for is not visible even to the mind's eye. What is visible is what corresponds to the characteristic flash of the diamond—it is something characteristic in the surroundings that betrays the presence of dog. It is not hard to see that "any dog" corresponds to any facet of the diamond; "my dog" corresponds to one particular facet of a diamond that is continually being "cut" by experience.

[TO BE CONTINUED.]

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